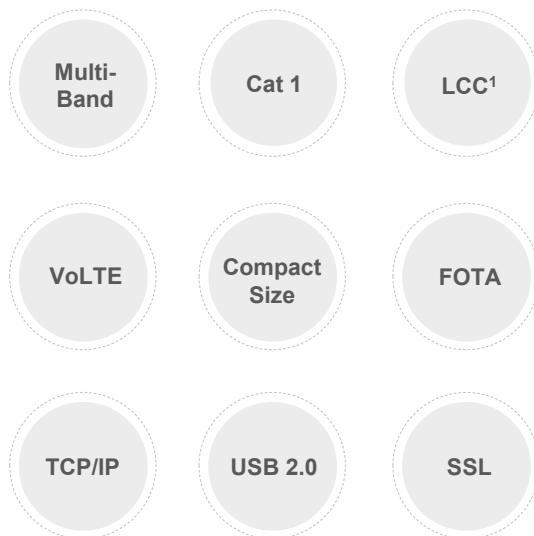
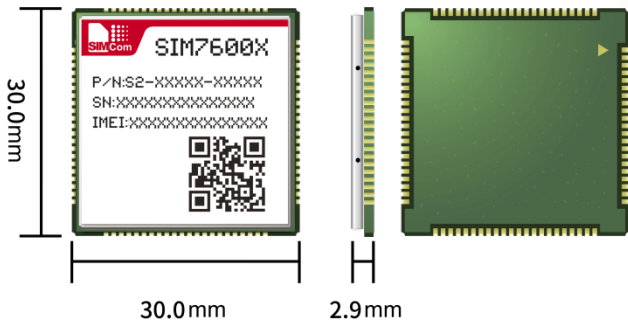


V: 2020.02

# SIM7600 Series

## SIMCom LTE Cat 1 Module



### Product Description

The SIM7600 series is the LTE Cat 1 module which supports wireless communication modes of LTE-TDD/LTE-FDD/HSPA+/GSM/GPRS/EDGE etc. It supports maximum 10Mbps downlink rate and 5Mbps uplink rate.

The SIM7600 series adopts LCC<sup>1</sup> form factor and is compatible with SIM5320/SIM5360 series(UMTS/HSPA+ modules), and SIM7600/SIM7600-H series(LTE modules), which enables smooth migration from 3G products to LTE products, and greatly facilitates more compatible product design for the customer needs.

The SIM7600 series integrates multiple satellite high accuracy positioning GNSS systems, with multiple built-in network protocols, supports drivers for main operation systems (USB driver for Windows, Linux and Android etc.) and software function, AT commands are compatible with SIM7500/SIM7600-H series modules. Meanwhile the SIM7600 series integrates main industrial standard interfaces, with powerful expansibility, including abundant interfaces such as UART, USB, I2C, GPIO, which is suitable for main IoT applications such as telematics, surveillance devices, CPE, industrial routers, and remote diagnostics etc.

### Key Benefits

- ◆ LCC<sup>1</sup> form factor with abundant interfaces
- ◆ Suitable for LTE, UMTS and GSM networks with global coverage
- ◆ Abundant software functions: FOTA, LBS, TLS, VoLTE
- ◆ Multi-constellation GNSS receiver support
- ◆ AT commands of the SIM7600 series are compatible with the SIM7500/SIM7600-H series modules

#### NOTE

1: The form factor of SIM7600NA & SIM7600G is LCC+LGA

| Product                |                    | SIM7600A                                                   | SIM7600SA                                              | SIM7600E                                                   | SIM7600E-L1C                                               |
|------------------------|--------------------|------------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|
| Form Factor            |                    | LCC, 87PIN                                                 | LCC, 87PIN                                             | LCC, 87PIN                                                 | LCC, 87PIN                                                 |
| Dimensions(mm)         |                    | 30.0*30.0*2.9                                              | 30.0*30.0*2.9                                          | 30.0*30.0*2.9                                              | 30.0*30.0*2.9                                              |
| Frequency Bands        | LTE-FDD            | B2/B4/B12                                                  | B1/B2/B3/B4/B5/B7/B8/B28/B66                           | B1/B3/B5/B7/B8/B20                                         | B1/B3/B7/B8/B20                                            |
|                        | LTE-TDD            |                                                            | B40                                                    | B38/B40/B41                                                |                                                            |
|                        | WCDMA              | B2/B5                                                      | B1/B2/B5/B8                                            | B1/B5/B8                                                   | B1/B8                                                      |
|                        | GSM                |                                                            | 850/900/1800/1900MHz                                   | 900/1800MHz                                                | 900/1800MHz                                                |
| GNSS                   |                    | o                                                          | o                                                      | o                                                          | o                                                          |
| Temperature            |                    | -40℃ ~ +85℃                                                | -40℃ ~ +85℃                                            | -40℃ ~ +85℃                                                | -40℃ ~ +85℃                                                |
| Electrical Features    |                    |                                                            |                                                        |                                                            |                                                            |
| Supply Voltage(V)      |                    | 3.4~ 4.2                                                   | 3.4~ 4.2                                               | 3.4~ 4.2                                                   | 3.4~ 4.2                                                   |
| Power Consumption (mA) | LTE                | 2.3                                                        | 2.3                                                    | 2.3                                                        | 2.3                                                        |
|                        | WCDMA, DRX=9       | 3.3                                                        | 3.3                                                    | 3.3                                                        | 3.3                                                        |
|                        | GSM, BS_PA_MFRMS=2 | 2.8                                                        | 2.8                                                    | 2.8                                                        | 2.8                                                        |
| Data Transfer          |                    |                                                            |                                                        |                                                            |                                                            |
| LTE(Mbps)              |                    | 10(DL)/5(UL)                                               | 10(DL)/5(UL)                                           | 10(DL)/5(UL)                                               | 10(DL)/5(UL)                                               |
| HSPA+(Mbps)            |                    | 42(DL)/5.76(UL)                                            | 42(DL)/5.76(UL)                                        | 42(DL)/5.76(UL)                                            | 42(DL)/5.76(UL)                                            |
| WCDMA(Kbps)            |                    | 384(DL)/384(UL)                                            | 384(DL)/384(UL)                                        | 384(DL)/384(UL)                                            | 384(DL)/384(UL)                                            |
| GPRS/EDGE(Kbps)        |                    |                                                            | 236.8(DL)/236.8(UL)                                    | 236.8(DL)/236.8(UL)                                        | 236.8(DL)/236.8(UL)                                        |
| Software Features      |                    |                                                            |                                                        |                                                            |                                                            |
| Protocol               |                    | TCP/IP/IPV4/IPV6/<br>Multi-PDP/FTP/FTPS/HTTP/<br>HTTPS/DNS | TCP/IP/IPV4/IPV6/Multi-PDP/<br>FTP/FTPS/HTTP/HTTPS/DNS | TCP/IP/IPV4/IPV6/<br>Multi-PDP/FTP/FTPS/<br>HTTP/HTTPS/DNS | TCP/IP/IPV4/IPV6/<br>Multi-PDP/FTP/FTPS/<br>HTTP/HTTPS/DNS |
| TLS                    |                    | •                                                          | •                                                      | •                                                          | •                                                          |
| File System            |                    | •                                                          | •                                                      | •                                                          | •                                                          |
| Audio Record/Play      |                    | o                                                          | o                                                      | o                                                          | o                                                          |
| TTS                    |                    | o                                                          | o                                                      | o                                                          | o                                                          |
| DTMF                   |                    | o                                                          | o                                                      | o                                                          | o                                                          |
| LBS                    |                    | •                                                          | •                                                      | •                                                          | •                                                          |
| FOTA                   |                    | •                                                          | •                                                      | •                                                          |                                                            |
| Jamming Detection      |                    | •                                                          | •                                                      | •                                                          | •                                                          |
| Android RIL            |                    | Android 5.0/6.0/7.0/8.0/9.0                                | Android 5.0/6.0/7.0/8.0/9.0                            | Android 5.0/6.0/7.0/8.0/9.0                                | Android 5.0/6.0/7.0/8.0/9.0                                |
| USB Driver             |                    | Microsoft Windows 7/8/10<br>Linux/Android                  | Microsoft Windows 7/8/10<br>Linux/Android              | Microsoft Windows 7/8/10<br>Linux/Android                  | Microsoft Windows 7/8/10<br>Linux/Android                  |
| WWAN (RNDIS)           |                    | •                                                          | •                                                      | •                                                          | •                                                          |
| ECM                    |                    | •                                                          | •                                                      | •                                                          | •                                                          |
| MBIM                   |                    | Win8/Win10                                                 | Win8/Win10                                             | Win8/Win10                                                 | Win8/Win10                                                 |
| NDIS                   |                    | Linux/Windows 7/8/10                                       | Linux/Windows 7/8/10                                   | Linux/Windows 7/8/10                                       | Linux/Windows 7/8/10                                       |
| Firmware Upgrade       |                    | USB/FOTA                                                   | USB/FOTA                                               | USB/FOTA                                                   | USB                                                        |
| Interfaces             |                    |                                                            |                                                        |                                                            |                                                            |
| SIM Card               |                    | 1.8V/3.0V                                                  | 1.8V/3.0V                                              | 1.8V/3.0V                                                  | 1.8V/3.0V                                                  |
| UART                   |                    | •                                                          | •                                                      | •                                                          | •                                                          |
| USB                    |                    | •                                                          | •                                                      | •                                                          | •                                                          |
| PCM                    |                    | o                                                          | o                                                      | o                                                          | o                                                          |
| ADC                    |                    | •                                                          | •                                                      | •                                                          | •                                                          |
| SD Card                |                    | •                                                          | •                                                      | •                                                          | •                                                          |
| SPI                    |                    | o                                                          | o                                                      | o                                                          | o                                                          |
| I2C                    |                    | •                                                          | •                                                      | •                                                          | •                                                          |
| Diversity Receiver     |                    | •                                                          | •                                                      | •                                                          | •                                                          |
| Certification          |                    |                                                            |                                                        |                                                            |                                                            |
| Regulatory             |                    | FCC/PTCRB/RoHS/REACH/IC                                    | CE-RED/ACMA/FCC/<br>RoHS/REACH/Anatel                  | CE-RED/RoHS/REACH/<br>IMDA/GCF                             | CE-RED/RoHS/REACH                                          |
| Carrier                |                    | AT&T/Rogers                                                | Telstra                                                | DTAG/Vodafone                                              |                                                            |

•: support o: optional \*: on going

| Product                |                   | SIM7600G                                                 | SIM7600NA                                          |
|------------------------|-------------------|----------------------------------------------------------|----------------------------------------------------|
| Form Factor            |                   | LCC+LGA, 119PIN                                          | LCC+LGA, 119PIN                                    |
| Dimensions(mm)         |                   | 30.0*30.0*2.9                                            | 30.0*30.0*2.9                                      |
| Frequency Bands        | LTE-FDD           | B1/B2/B3/B4/B5/B7/B8/B12/B13/B18/B19/B20/B25/B26/B28/B66 | B2/B4/B5/B12/B13/B14/B25/B26/B66/B71               |
|                        | LTE-TDD           | B34/B38/B39/B40/B41                                      | B41                                                |
|                        | WCDMA             | B1/B2/B4/B5/B6/B8/B19                                    |                                                    |
|                        | GSM               | 850/900/1800/1900MHz                                     |                                                    |
| GNSS                   |                   | o                                                        | o                                                  |
| Temperature            |                   | -40℃ ~ +85℃                                              | -40℃ ~ +85℃                                        |
| Electrical Features    |                   |                                                          |                                                    |
| Supply Voltage(V)      |                   | 3.4~ 4.2                                                 | 3.4~ 4.2                                           |
| Power Consumption (mA) | LTE               | 2.3                                                      | 2.3                                                |
|                        | WCDMA, DRX=9      | 3.3                                                      |                                                    |
|                        | GSM,BS_PA_MFRMS=2 | 2.8                                                      |                                                    |
|                        |                   |                                                          |                                                    |
| Data Transfer          |                   |                                                          |                                                    |
| LTE(Mbps)              |                   | 10(DL)/5(UL)                                             | 10(DL)/5(UL)                                       |
| HSPA+(Mbps)            |                   | 42(DL)/5.76(UL)                                          |                                                    |
| WCDMA(Kbps)            |                   | 384(DL)/384(UL)                                          |                                                    |
| GPRS/EDGE(Kbps)        |                   | 236.8(DL)/236.8(UL)                                      |                                                    |
| Software Features      |                   |                                                          |                                                    |
| Protocol               |                   | TCP/IP/IPV4/IPV6/Multi-PDP/FTP/FTPS/HTTP/HTTPS/DNS       | TCP/IP/IPV4/IPV6/Multi-PDP/FTP/FTPS/HTTP/HTTPS/DNS |
| TLS                    |                   | •                                                        | •                                                  |
| File System            |                   | •                                                        | •                                                  |
| Audio Record/Play      |                   | o                                                        | o                                                  |
| TTS                    |                   | o                                                        | o                                                  |
| DTMF                   |                   | o                                                        | o                                                  |
| LBS                    |                   | •                                                        | •                                                  |
| FOTA                   |                   | •                                                        | •                                                  |
| Jamming Detection      |                   | •                                                        | •                                                  |
| Android RIL            |                   | Android 5.0/6.0/7.0/8.0/9.0                              | Android 5.0/6.0/7.0/8.0/9.0                        |
| USB Driver             |                   | Microsoft Windows 7/8/10<br>Linux/Android                | Microsoft Windows 7/8/10<br>Linux/Android          |
| WWAN (RNDIS)           |                   | •                                                        | •                                                  |
| ECM                    |                   | •                                                        | •                                                  |
| MBIM                   |                   | Win8/Win10                                               | Win8/Win10                                         |
| NDIS                   |                   | Linux/Windows 7/8/10                                     | Linux/Windows 7/8/10                               |
| Firmware Upgrade       |                   | USB/FOTA                                                 | USB/FOTA                                           |
| Interfaces             |                   |                                                          |                                                    |
| SIM Card               |                   | 1.8V/3.0V                                                | 1.8V/3.0V                                          |
| UART                   |                   | •                                                        | •                                                  |
| USB                    |                   | •                                                        | •                                                  |
| PCM                    |                   | o                                                        | o                                                  |
| ADC                    |                   | •                                                        | •                                                  |
| SD Card                |                   | •                                                        | •                                                  |
| SPI                    |                   | o                                                        | o                                                  |
| I2C                    |                   | •                                                        | •                                                  |
| SGMII                  |                   | o                                                        | o                                                  |
| Diversity Receiver     |                   | •                                                        | •                                                  |
| Certification          |                   |                                                          |                                                    |
| Regulatory             |                   | CE-RED/FCC/IC/RoHS*/REACH*                               | FCC*/IC*/PTCRB*/GCF*/RoHS*/REACH*                  |
| Carrier                |                   |                                                          | AT&T*/Verizon*/Sprint*/T-mobile*                   |

•: support o: optional \*: on going